

Structured Computer Architecture

Chapter 3.14: Parallel Computing

N-Order Systems (N-OS)

Based on Structured Computer Architecture: An Order-Based Approach

Outline

Learning Objectives

By the end of this chapter, you will understand:

- ▶ N-Order Systems (N-OS) and their theoretical foundations
- ▶ Cellular Automata concepts and applications
- ▶ Controlled Cellular Automata (CCA) architecture
- ▶ Parallel computing paradigms and mapping operations
- ▶ Distribution networks and scalability challenges
- ▶ Space-time trade-offs in parallel computation
- ▶ Moore's Law implications for parallel systems

From Sequential to Parallel: The N-OS Evolution

Motivation for N-Order Systems:

- ▶ Previous chapters: Sequential feedback loops (1-OS to 5-OS)
- ▶ Question: Can we continue closing new loops?
- ▶ Goal: Additional gains from additional loops
- ▶ Challenge: Managing complexity as N increases

Complexity Management:

- ▶ **Size:** Proportional to N (acceptable)
- ▶ **Complexity:** Must remain independent of N
- ▶ **Solution:** Loops close over identical systems
- ▶ **Repetitive patterns:** Keep description controllable

Moore's Law Constraint:

- ▶ Exponential growth in size (transistors)

System Order Evolution

System Evolution

0-OS: Combinational
~~1-OS: Memory~~
~~2-OS: Automata~~
~~3-OS: Processors~~
~~4-OS: Von Neumann~~
~~5-OS: Harvard~~
N-OS: Parallel